NOETHERIAN FILTERED COLIMITS OF ETALE ALGEBRAS HAVE FINITELY MANY SEPARABLE FIBRES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- An etale algebra over a field is a finite product of finite separable field extensions. the author cites Algebra, Lemma algebra-lemma-etale-over-field at proof lines 11251-11253. Identifies the pieces of the colimit and gives the separability.
- A Noetherian ring with finitely many primes and no nontrivial specialisations splits as a finite product of fields. the author cites Algebra, Lemma algebra-lemma-total-ring-fractions-no-embedded-points and Algebra, Proposition algebra-proposition-dimension-zero-ring at proof lines 11260-11264. Produces the product decomposition of the fibre.

Source reference. Original lemma 0AH1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13214–13214. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 00U3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1097–1097. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 02LX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3424–3424. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00KJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 753–753. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let A be a ring. Let B be a filtered colimit of étale A-algebras. Let $\mathfrak{p}$ be a prime of A. If B is Noetherian, then there are finitely many primes $\mathfrak{q}_1, \ldots, \mathfrak{q}_r$ lying over $\mathfrak{p}$, we have $B \otimes_A \kappa(\mathfrak{p}) = \prod \kappa(\mathfrak{q}_i)$, and each of the field extensions $\kappa(\mathfrak{q}_i)/\kappa(\mathfrak{p})$ is separable algebraic.*

Proof. Write B as a filtered colimit $B = \operatorname{colim} B_i$ with $A \to B_i$ étale. Then on the one hand $B \otimes_A \kappa(\mathfrak{p}) = \operatorname{colim} B_i \otimes_A \kappa(\mathfrak{p})$ is a filtered colimit of étale $\kappa(\mathfrak{p})$-algebras, and on the other hand it is Noetherian. An étale $\kappa(\mathfrak{p})$-algebra is a finite product of finite separable field extensions (Algebra, Lemma 00U3). Hence there are no nontrivial specializations between the primes (which are all maximal and minimal primes) of the algebras $B_i \otimes_A \kappa(\mathfrak{p})$ and hence there are no nontrivial specializations between the primes of $B \otimes_A \kappa(\mathfrak{p})$. Thus $B \otimes_A \kappa(\mathfrak{p})$ is reduced and has finitely many primes which all minimal. Thus it is a finite product of fields (use Algebra, Lemma 02LX or Algebra, Proposition 00KJ). Each of these fields is a colimit of finite separable extensions and hence the final statement of the lemma follows. □